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Engineering Drought Resilience Through Plant Architecture in Tepary Beans

Abstract

Drought stress is a major constraint on agricultural productivity, yet the relationships among plant architecture and water-use efficiency (WUE) remain poorly understood. In particular, **it is unclear which shoot topologies promote drought resilience under water-limited conditions**. This study investigated how architectural and developmental modifications influence physiological responses to drought in tepary bean (*Phaseolus acutifolius*) and soybean (*Glycine max*). In the first tepary bean experiment, stomatal conductance was compared between the highest and lowest mature leaves under well-watered and drought conditions to evaluate the spatial distribution of stomatal conductance across the shoot and its relationship to physiological water allocation. A second tepary bean experiment investigated hormone-mediated architectural regulation using paclobutrazol (PAC), indole-3-acetic acid (IAA), and control treatments under well-watered and drought conditions. A complementary soybean experiment evaluated the effects of shoot apical meristem decapitation under well-watered and drought conditions. Across the hormone-treated tepary and soybean experiments, the PhenoSight imaging platform quantified plant architecture, chlorophyll fluorescence, and Photosystem II (PSII) efficiency, while stomatal conductance was measured using an SC-1 Leaf Porometer. Preliminary results indicated that **compact and prostrate tepary architectures exhibited enhanced drought performance relative to more erect growth forms**. PAC promoted increased plant compactness and increased allocation of water loss in the highest mature leaves during drought. Future work will integrate Rapid Anatomy Tools (RAT), CPlantBox modeling of radial and axial water flow, and machine learning-assisted cross-sectional image analysis to identify anatomical and hydraulic traits associated with improved drought resilience and water-use efficiency.